

The Finite-Defect Numerical-Range Conjecture Is Affirmative

Literature answer to arXiv:1009.2249 via arXiv:1205.2025

12 August 2026

Status: fully answered in later literature. Bercovici and Timotin prove the economical unitary-dilation intersection formula for every contraction whose two defect indices are equal and finite. Their stronger support-line theorem also proves the source conjecture's full “wrapping” assertion.

The source conjecture

For a contraction $T \in \mathcal{L}(\mathcal{H})$, let

$$D_T = (I - T^*T)^{1/2}, \quad \mathcal{D}_T = \overline{D_T \mathcal{H}}.$$

Assume $\dim \mathcal{D}_T = \dim \mathcal{D}_{T^*} = N < \infty$, and let $\mathcal{U}$ be the set of unitary N -dilations of T acting on $\mathcal{H} \oplus \mathbb{C}^N$. If $W(A) = \{\langle Ax, x \rangle : \|x\| = 1\}$ denotes numerical range, the source paper defines $\tau(U) = \overline{W(U)}$ and says that τ *wraps* $\overline{W(T)}$ when every open half-plane containing $\overline{W(T)}$ contains $\overline{W(U)}$ for some $U \in \mathcal{U}$. Conjecture 6.1 asks whether this always holds. In particular it asks for

$$\overline{W(T)} = \bigcap_{U \in \mathcal{U}} \overline{W(U)}. \tag{1}$$

Exact later answer

Two years later, Hari Bercovici and Dan Timotin proved the following. Theorem 2.4 of arXiv:1205.2025 assumes precisely the same condition $\dim \mathcal{D}_T = \dim \mathcal{D}_{T^*} = n < \infty$:

1. every support line d of $\overline{W(T)}$ is also a support line of $\overline{W(U)}$ for some unitary n -dilation U of T ; and
2. the intersection identity (1) holds using only those economical n -dilations.

Why the stronger wrapping clause also follows

The later paper explicitly states the intersection formula, while the source conjecture asks for the formally stronger wrapping property. Theorem 2.4(1) bridges the two directly.

Let G be an open half-plane containing the compact convex set $A = \overline{W(T)}$. After rotating coordinates, write

$$G = \{z : \operatorname{Re} z < c\}, \quad a = \max_{z \in A} \operatorname{Re} z < c.$$

The scalar case of Theorem 5.3 is contained in [11, Theorem 6.3].

6. Final remarks

It seems natural to formulate the following conjecture, which would complement Choi and Li's answer to Halmos' question.

Conjecture 6.1. *Suppose $T \in \mathcal{L}(\mathcal{H})$ is a contraction with $\dim \mathcal{D}_T = \dim \mathcal{D}_{T^*} = N < \infty$, $\mathcal{U}$ is the set of unitary N -dilations of T to $\mathcal{H} \oplus \mathbb{C}^N$, and $\tau : \mathcal{U} \rightarrow \mathfrak{K}$ is defined by $\tau(\mathbf{U}) = \overline{W(\mathbf{U})}$. Then τ wraps $\overline{W(T)}$. In particular,*

$$\overline{W(T)} = \bigcap_{\mathbf{U} \in \mathcal{U}} \overline{W(\mathbf{U})} \quad (6.1)$$

Note that the conjecture is open even for $N = 1$. The main points that have been settled are presented below. In the sequel $T \in \mathcal{L}(\mathcal{H})$ will be a contraction with $\dim \mathcal{D}_T = \dim \mathcal{D}_{T^*} = N < \infty$.

6.1. Theorem 4.8 shows that the conjecture is true for $C_0(N)$ contractions.

6.2. As we show below, if we add a unitary operator to one for which the conjecture holds, the conjecture will still hold.

Lemma 6.2. *If $T_1 = T \oplus V$, where V is unitary, and Conjecture 6.1 is true for T , then it is true for T_1 .*

Proof. The unitary N -dilations of $T \oplus V$, for V unitary, are exactly $\mathbf{U} \oplus V$, with $\mathbf{U}$ a unitary N -dilation of T . Since the numerical range of a direct sum is the convex hull of the numerical ranges of the components, the statement follows from Lemma 4.3. $\square$

Figure 1: Conjecture 6.1 in Benhida–Gorkin–Timotin, arXiv:1009.2249, PDF page 12.

The line $d = \{z : \operatorname{Re} z = a\}$ supports A . Theorem 2.4(1) provides a unitary n -dilation U for which d supports $\overline{W(U)}$. Since compression gives $A \subseteq \overline{W(U)}$, both sets lie on the same side of d . Consequently

$$\overline{W(U)} \subseteq \{z : \operatorname{Re} z \leq a\} \subset G.$$

This is exactly the source definition of wrapping. Thus Conjecture 6.1 is affirmative in its full stated strength, for every finite N and every contraction with equal defect indices.

Scope match

There is no loss of hypotheses: both papers concern contractions on complex separable Hilbert spaces, equal finite defect indices, and unitary dilations on $\mathcal{H} \oplus \mathbb{C}^N$. The later theorem covers arbitrary contractions in this class, not merely the $C_0(N)$ case proved in the source paper.

Lemma 2.3. *Assume that U is a unitary n -dilation of T , $\lambda_1, \dots, \lambda_{n+1}$ are eigenvalues of U (not necessarily distinct, but with repeated values being multiple eigenvalues). Then the convex hull of $\lambda_1, \dots, \lambda_{n+1}$ contains a point in $W(T)$.*

Proof. If $\xi_1, \dots, \xi_{n+1}$ are corresponding eigenvectors of U , then their linear span Y has dimension $n+1$, and thus its intersection with $\mathcal{H}$ contains a vector ξ of norm 1. Then $\langle U\xi, \xi \rangle = \langle T\xi, \xi \rangle$ belongs to the numerical range of the restriction of U to Y , which is precisely the convex hull of $\lambda_1, \dots, \lambda_{n+1}$. $\square$

Theorem 2.4. *Assume that T is a contraction with $\dim \mathcal{D}_T = \dim \mathcal{D}_{T^*} = n < \infty$.*

- (1) *If d is a support line for $\overline{W(T)}$, then there exists a unitary n -dilation U of T such that d is a support line for $\overline{W(U)}$.*
- (2) *$\overline{W(T)} = \bigcap \{\overline{W(U)} : U \in \mathcal{L}(\mathcal{K}) \text{ is a unitary } n\text{-dilation of } T\}$.*

Proof. Obviously (2) follows from (1), so we only have to prove (1). Assertion (1) is immediate if d is tangent to $\mathbb{T}$, so we may assume that d intersects $\mathbb{T}$ in two points. These two points must belong to the closure of one of the component arcs $\widehat{z_1, z_2}$ of $\mathbb{T} \setminus \sigma(T) = \mathbb{T} \setminus \overline{W(T)}$.

Consider then, for $\epsilon > 0$, the line d_ϵ parallel to d at a distance ϵ and that does not meet $W(T)$. For ϵ sufficiently small, d_ϵ intersects $\widehat{z_1, z_2}$ in two points $\lambda_\epsilon, \lambda'_\epsilon$. By Proposition 2.2 there exists a unitary n -dilation U_ϵ of T (acting on $\mathcal{H} \oplus \mathcal{D}_T$) which has λ_ϵ as an eigenvalue of multiplicity n . We assert that d is a support line for $W(U_\epsilon)$. Indeed, if U had an eigenvalue λ'_ϵ on $\mathbb{T} \cap \widehat{z_1, z_2}$, we could apply Lemma 2.3 to the set formed by λ repeated n times together with λ' to obtain an element in $W(T)$ on the segment $[\lambda_\epsilon, \lambda'_\epsilon]$ which is at distance ϵ from $W(T)$.

Finally, noting that the set of unitary n -dilations of T on $\mathcal{H} \oplus \mathcal{D}_T$ is compact, a limit point of the unitary dilations U_ϵ will itself have d as a support line. $\square$

Note that based on the Sz.-Nagy—Foias functional model [27] one can give a function theoretical description of all unitary n -dilations of T ; this description,

Figure 2: Theorem 2.4 in Bercovici–Timotin, arXiv:1205.2025, PDF page 4.

References

- [1] C. Benhida, P. Gorkin, and D. Timotin, *Numerical ranges of $C_0(N)$ contractions*, arXiv:1009.2249; Integral Equations Operator Theory 70 (2011), 265–279.
- [2] H. Bercovici and D. Timotin, *The numerical range of a contraction with finite defect numbers*, arXiv:1205.2025; J. Math. Anal. Appl. 417 (2014), 42–56.